Biologically informed neural network identifies Unfolded Protein Response as key pathway in critical COVID-19

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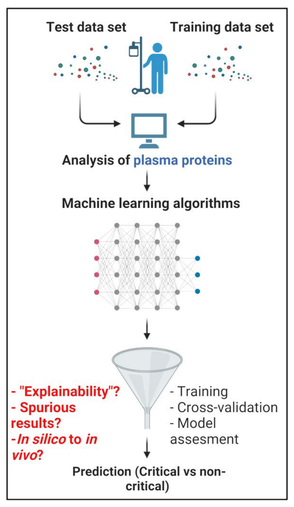
Highlights

✓**COV-PASnet** is a ***biologically-informed neural network*** receiving as input COVID-19 plasma proteomics and biological pathways

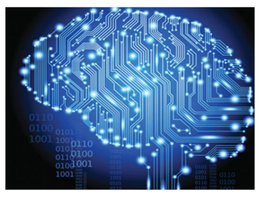
✓COV-PASnet unveils **Unfolded Protein Response (UPR)** as previously unexplored predictive pathway in critical COVID-19

✓**UPR** is confirmed in **plasmablasts during critical COVID-19**, paving the way for potential therapeutic interventions.

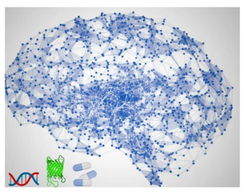
Background-Objective



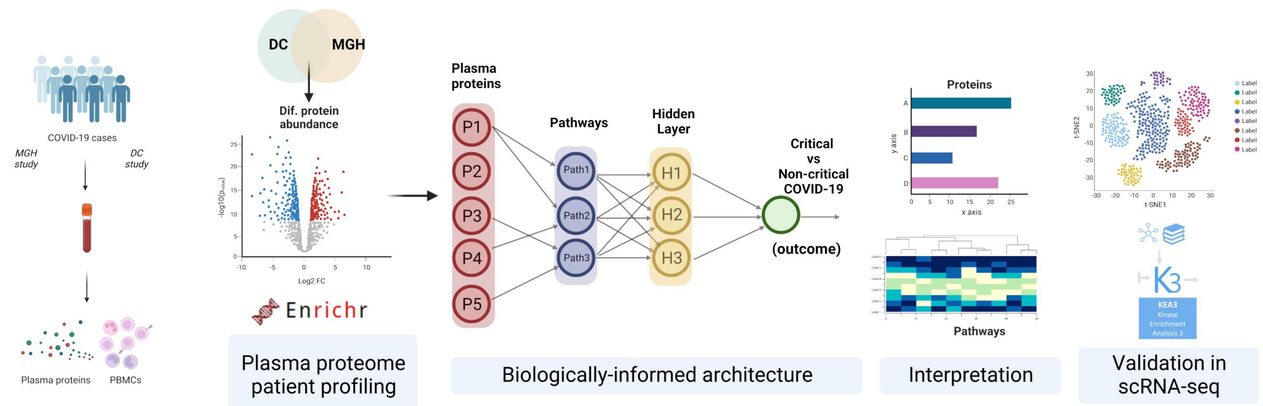
Objective: "Educate" an "interpretable" AI tool to find novel biomarkers and druggable targets in COVID-19



Background: Harnessing Artificial Intelligence (AI) for -omics is key to discover new actionable knowledge but there are caveats [1]!

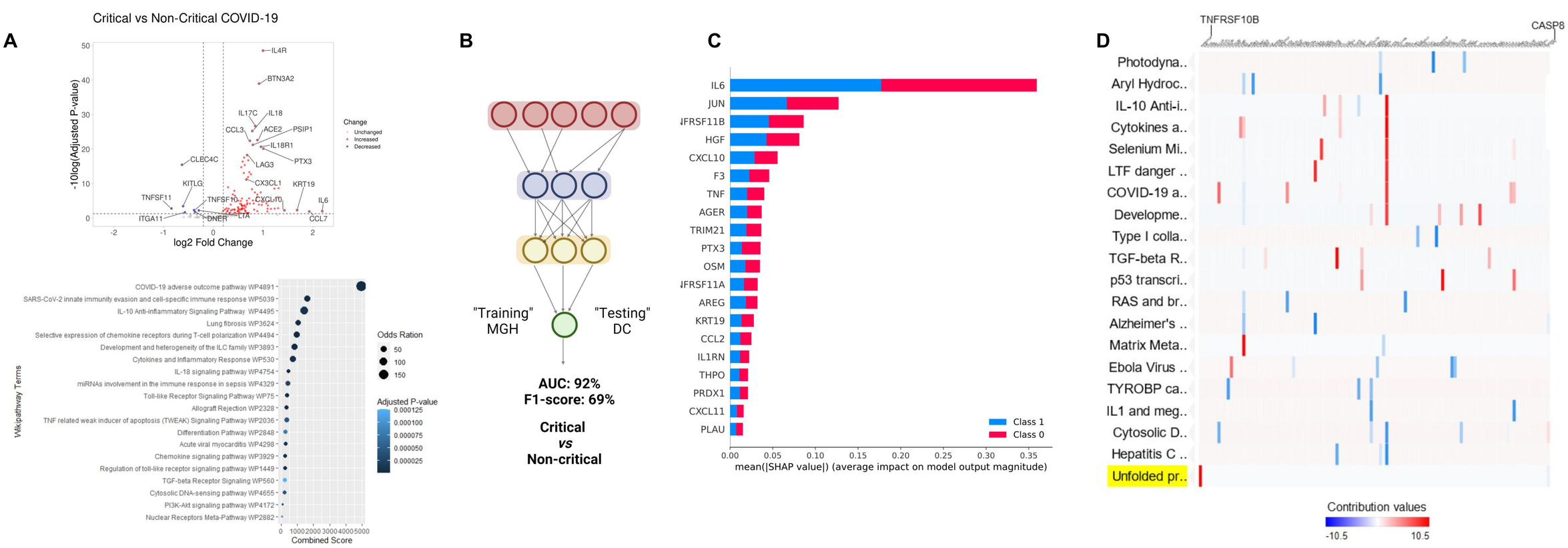


Methods

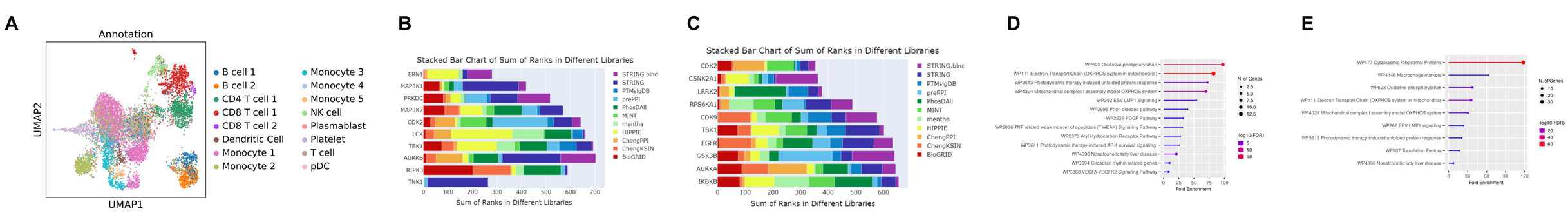


COV-PASnet, a biologically-informed neural network (NN) to demarcate critical COVID-19 cases. COV-PASnet is a pathway-associated sparse deep NN with explainable Artificial Intelligence Shapley values [2]. **Inputs**: plasma proteins from MGH/DC datasets [3,4] and pathways (Enrichr). **Outputs:** Association of proteins-pathways, with predictive scoring. **Biological validation:** Kinase enrichment analysis (KEA3) on differentially expressed genes for each scRNA-seq cell type.

Results



1) COV-PASnet unveils novel prognostic pathways in critical COVID-19 plasma proteome. (A) Inputs of COV-PASnet, i.e. differentially abundant proteins in critical COVID-19 for Day 0 (upper volcano plot) and significant pathways (lower bubble plot). **(B)** Training, testing and robustness of COV-PASnet in demarcating critical COVID-19 cases. **(C)** Top-predictive proteins (SHAP values barplot). **(D)** Unfolded Protein Response (UPR) which controls cellular proteostasis emerging as a top-predictive pathway in critical COVID-19 based on **TNFRSF10** and **CASP8** (heatmap).



2) Biological validation of COV-PASnet by detecting UPR-skewed plasmablasts in critical COVID-19. (A) UMAP clustering of **PBMC scRNA-seq data** from MGH-subcohort. **(B-C)** Stacked barplots from KEA3 showing **ERN1 (IRE1), a major UPR regulator** as the most inferred kinase in critical COVID-19 (B) and proliferative kinases in non-critical COVID-19 (C). **(D-E)** Barplots showing pathway enrichment (Wikipathways) for critical and non-critical COVID-19 plasmablasts where UPR is more prominent in the severe COVID-19 cases (D).

Conclusion

1. **COV-PASnet is a biologically-informed neural network that robustly demarcates *critical COVID-19* cases from early-samples of plasma proteins**
2. **COV-PASnet reveals known and unexplored prognostic pathways -one of the latter is UPR which modulates proteostasis [5]!**
3. **UPR hyper-activation through DR5 is an apoptotic trigger [6]-since plasmablasts participate in antibody secretion during viral infection [7], our findings merit further investigation to reveal whether UPR is a potential "druggable" target.**

Bibliography

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